FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
CONCEPT RECAPITULATION TEST – III
PAPER –1
TEST DATE: 01-05-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

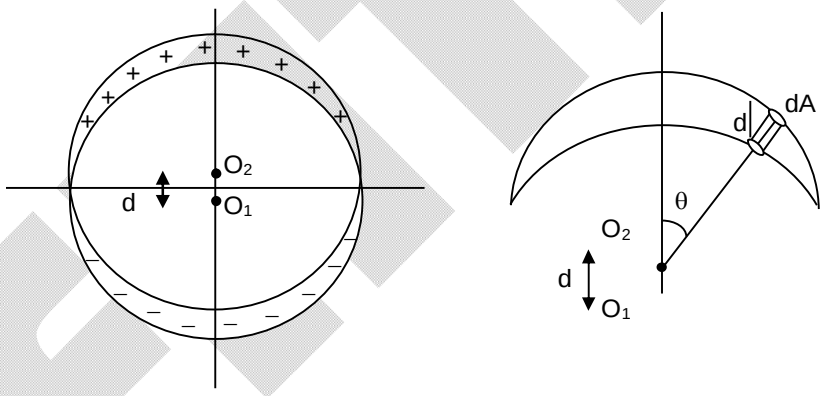
1. BCD

Sol. Net field inside = 0

$E^{ind} = E_0(\downarrow)$ down word direction and uniform

Thus surface change density $\sigma = \sigma_0 \cos\theta$

We can think two uniformly charged sphere of density ρ and $-\rho$ separated by a very small distance d.



$dq = \rho.d\cos\theta \cdot dA$

$\Rightarrow \quad \sigma = \rho d.\cos\theta = \sigma_0 \cos\theta$

$\Rightarrow \quad$ Now $E^{ind} = \dfrac{\rho}{3\epsilon_0}.d$

$\Rightarrow \quad \sigma_0 = 3\epsilon_0 E_0$

Field near surface $E = \dfrac{\sigma}{\epsilon_0} = 3E_0 \cdot \cos\theta$

Dipole moment

$p = \rho.\dfrac{4}{3}\pi R^3 \cdot d = 4\pi\epsilon_0 E_0 R^3$

2. ACD

Sol. Uniform ring

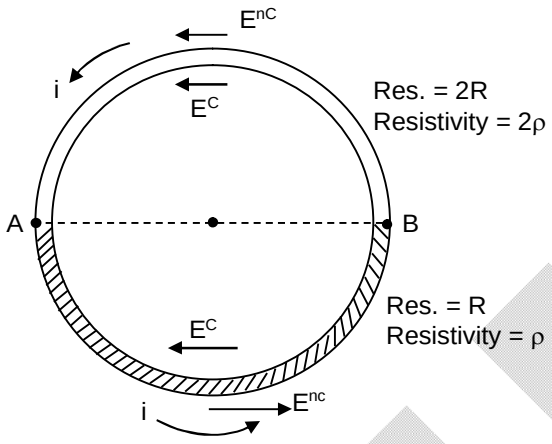
$E^{nc} \cdot 2\pi a = \pi a^2 \cdot \frac{d}{dt}(kt) \Rightarrow E^{nc} = \frac{ak}{2}$ (Anticlockwise)

$E^c = 0, \qquad V_{AB} = 0$

Non-uniform ring

$i = \frac{\pi a^2 k}{3R}$ (same current)

$J = \frac{1}{\rho}E = \text{same} \Rightarrow \frac{E}{\rho} = \text{cont.}$



Thus $\frac{E^{nC} + E^c}{E^{nc} - E^c} = 2 \Rightarrow E^c = \frac{E^{nc}}{3} = \frac{ak}{6}$

$V_{BA} = \frac{ak}{6} \times \pi a = \frac{\pi k a^2}{6}$

3. BD

Sol. Cycloidal path

$R = \frac{mv}{qB}$ whose $v = \frac{E}{B} = 10^3 m/s$

$= \frac{10^{-9} \times 10^3}{2 \times 10^{-6} \times 5 \times 15^3} = 100$ m

Max. Speed $2v = 2 \times 10^3 m/s$

$\vec{L} = 2R \cdot m(2v)\left(-\hat{k}\right)$

$= 4 \times 10^{-9} \times 10^3 \times 10^2 = 4 \times 10^{-4} kg\, m/s \left(-\hat{k}\right)$

4. C

Sol. $L = \frac{2h}{\pi} \Rightarrow n = 4$

$r_n = a_0 \frac{n^2}{Z} \Rightarrow 4a_0 = \frac{a_0 (4)^2}{Z} Z = 4$

Now six wavelengths can be possible as

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = 16R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

5. B

Sol. For B = 0,

$$i_c = i_d$$

$$\Rightarrow \quad \epsilon_0 \frac{d\phi_E}{dt}$$

$$\Rightarrow \quad I = \epsilon_0 \cdot \frac{d}{dt}\left(Kt \cdot \pi a^2\right) = \pi a^2 \epsilon_0 K$$

$$\Rightarrow \quad K = \frac{I}{\pi a^2 \epsilon_0}$$

6. A

Sol. $\ell_1 = \frac{\lambda}{4}, \ \ell_2 = \frac{3\lambda}{4}$ thus $\lambda = 2(\ell_2 - \ell_1)$

$v = f\lambda = 2 \times 150 \times 100 \times 10^{-2} = 300 \text{ m/s}$

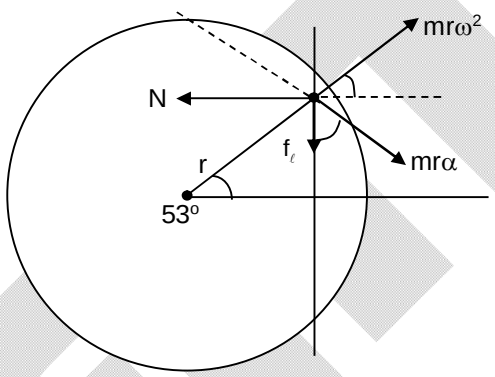
$$\frac{\Delta v}{v} = \frac{\Delta\ell_1 + \Delta\ell_2}{\ell_2 - \ell_1} = \frac{0.5 + 0.1}{100} = \frac{0.6}{100}$$

%Error is 0.6%

7. B

Sol. t = 0, $\omega = 0$

Pulling force = $\frac{3}{5}mr\alpha$



$$f_\ell = 0.8 \times \frac{4}{5}mr\alpha$$

$f_\ell >$ Pulling force

Body at rest at t = 0

At limiting condition

$$f_\ell + \frac{3}{5}mr\alpha = \frac{4}{5}mr\omega^2 \quad \text{and} \quad N = \frac{3}{5}mr\omega^2 + \frac{4}{5}mr\alpha$$

$$\Rightarrow \quad 0.8 \times \left[\frac{3}{5}mr\omega^2 + \frac{4}{5}mr\alpha\right] + \frac{3}{5}mr\alpha = \frac{4}{5}mr\omega^2$$

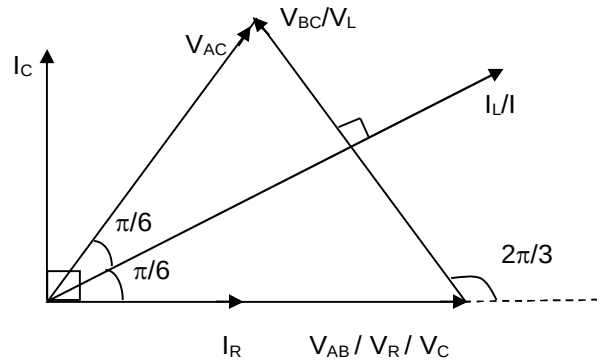
$$\Rightarrow \quad \frac{4}{5} \times \frac{3}{5}\omega^2 + \frac{4}{5} \times \frac{4}{5}\alpha + \frac{3}{5}\alpha = \frac{4}{5}\omega^2$$

$$\Rightarrow \quad \frac{31}{25}\alpha = \frac{8}{25}\omega^2 \Rightarrow 31\times 2 = 8\times 4t^2$$

$$\Rightarrow \quad t = \sqrt{\frac{31}{16}}\,\text{sec}$$

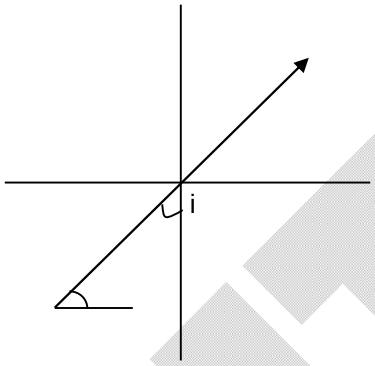
8. B

Sol.



9. D

Sol. By shell's Law:



$\mu \sin i$ = constant

= 1 here

$\therefore \mu = \sec\theta$

$$= \sqrt{1+\left(\frac{dy}{dx}\right)^2}$$

Case 1: $\frac{dy}{dx}$ changes from O to ∞ as y changes from O to R

Hence the circle must be $x^2 + (y-R)^2 = R^2$

$$\frac{dy}{dx} = \frac{x}{R-y}$$

$$\therefore \mu = \frac{R}{R-y} = \frac{4}{3}\text{at} = \frac{R}{4}$$

$$\text{If } y = \frac{2R}{5}, x^2 + \left(\frac{2R}{5} - R\right)^2 = R^2 \Rightarrow x = \frac{4R}{5}$$

Case 2: The circle must be $(x-R)^2 + y^2 = R^2$

If $y=\frac{3R}{5},(x-R)^2+\left(\frac{3R}{5}\right)^2=R^2$

$\therefore x=\frac{R}{5}$

Here, $\frac{dy}{dx}=\frac{R-x}{y}$

10. D

Sol. Chang in entropy

$\int_{S_1}^{S_2} dS=\int\frac{dQ}{T}$

$\Delta S=(S_2-S_1)=\int\frac{dQ}{T}$

$f_{eq}=\left(\frac{n_1f_1+n_2f_2}{n_1+n_2}\right)=\frac{(1\times3+1\times5)}{2}=4$

For $A\rightarrow B: TV^{1/2}=C_1$

$\frac{PV^{3/2}}{nR}=C_1$

$PV^{3/2}=C_2$

$C=C_v+\frac{R}{1-x}=\left(\frac{4R}{2}\right)+\frac{R}{1-\frac{1}{2}}=(2R-2R)=0$

So, $dQ=nCdT=0$

$\Delta S=0$ So, $S_1=S_2$

B to C : T = constant and B to C it is case of compression so dW is (-Ve) and dQ = dW +dU dQ will be (-ve) means dS is (-ve)

$S_2<S_1$

C to D : Isochoric process dW = 0

$dU=-ve; dQ=-ve$

So, $\Delta s=-ve$

$S_2<S_1$

D to A : TV = constant

$\frac{PV}{nR}V=$ constnat

$PV^2=C$

$C=\left(\frac{4R}{2}\right)+\left(\frac{R}{1-2}\right)=2R-R=R$

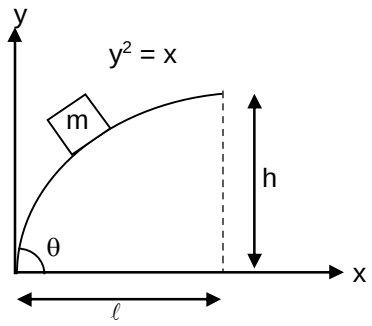
$dQ=nCdT<0$

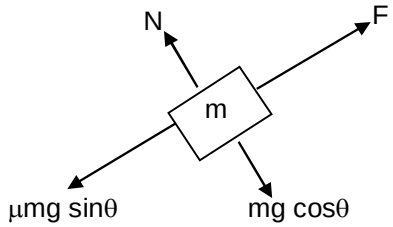
So, $dS<0$; So, $S_2<S_1$

Because (dT < 0)

11. A

Sol.





$w_r = \mu mg\ell$

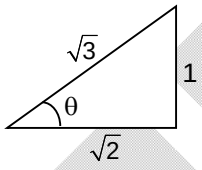
$w_f = 10J$

$\tan\theta = \frac{dy}{dx}$ $\quad 2y\frac{dy}{dx} = 1$

$\tan\theta = \frac{1}{2y}$ $\quad \tan\theta = \frac{1}{2\sqrt{x}}$

$\tan\theta = \frac{1}{\sqrt{2}}$, at $x = \frac{1}{2}$

$N = mg\cos\theta$



$N = 2\times10\times\sqrt{\frac{2}{3}} = 20\sqrt{\frac{2}{3}}N$

$w_f = \mu mg\ell + mgh$

$w_f = 0.5\times20\times1 + 20\times1$

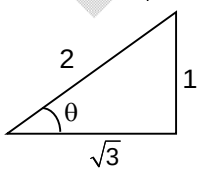
$w_f = 30J$

At any point : $N = mg\cos\theta$

$f = \mu mg\cos\theta$

$R = \sqrt{N^2 + f^2}$

$R = mg\cos\sqrt{1+\mu^2}$



At $x = \frac{3}{4}; \tan\theta = \frac{1}{\sqrt{3}}$

$R = 20 \times \frac{\sqrt{3}}{2} \times \sqrt{1+\frac{1}{4}} = 5\sqrt{15}N$

SECTION – B

12. 5

Sol. $10\frac{\lambda D}{d} = 2.5\times10^{-3}m \quad \Rightarrow \quad d = \frac{15}{2.5}\times10^{-3}m$

$y = 0.1mm, \quad \Delta\phi = \frac{2\pi}{\lambda}\cdot\frac{yd}{D} = \frac{4\pi}{5}$

$I = I_{max}\cdot\cos^2\frac{\Delta\phi}{2}$

13. 4

Sol. $I = \frac{mh^2}{6}$

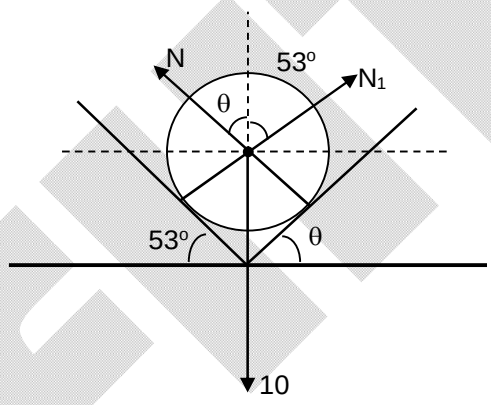
Same mass, $I \propto h^2$

$\frac{I_1}{I_2} = 4$

14. 8

Sol. $\frac{4N_1}{5} = N\sin\theta$ …(i)

$10 = N\cos\theta + \frac{3N_i}{5}$ …(ii)



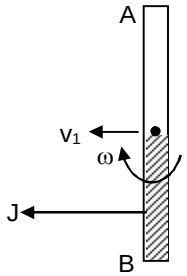
Now $N = \frac{10}{\left(\frac{3}{4}\sin\theta + \cos\theta\right)}$

$N_{min} = 8N$

15. 15

Sol. Impact will be at middle of half rods.

Just after



$J = m(v_1 + v)$ ...(i)

$J \cdot \frac{L}{4} = \frac{mL}{12}\omega$ ...(ii)

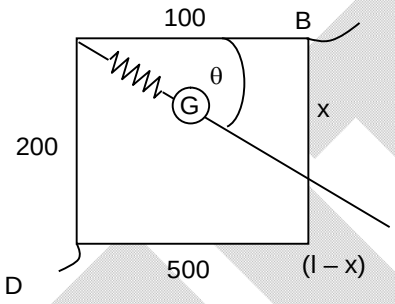
$e = \frac{1}{2} = \frac{v_1 + \frac{L\omega}{4}}{v_0}$ ...(iii)

Solving $v_1 = -\frac{v_0}{7}$, $\omega = \frac{18v_0}{7L}$

$v_B = \frac{L\omega}{2} - v_1 = \frac{8v_0}{7}$

16. 74

Sol. $\frac{100}{\left(\frac{400x}{\ell}\right)} = \frac{200}{\left[500 + \frac{400(\ell - x)}{\ell}\right]}$



$\Rightarrow \quad \frac{x}{\ell} = \frac{3}{4}$

$\Rightarrow \quad \theta = 37^0, t = 74\,\text{sec onds}$

17. 250

Sol. $R_{eff} = \frac{6R}{6+R} + 18 = \frac{24R + 108}{(6+R)}$

Power through 'R' is maximum

$\frac{d}{dR}\left[\left(\frac{20}{R_{eff}} \times \frac{6}{(6+R)}\right)^2 \cdot R\right] = 0$

$\Rightarrow \quad R = \frac{9}{2}\Omega$

$V_c = i_R \times R = 2.5\text{volt}$

$q = 250\mu C$

Chemistry

PART – II

SECTION – A

18. BC

Sol. pH of HCl cannot be more than 7 at 25^oC

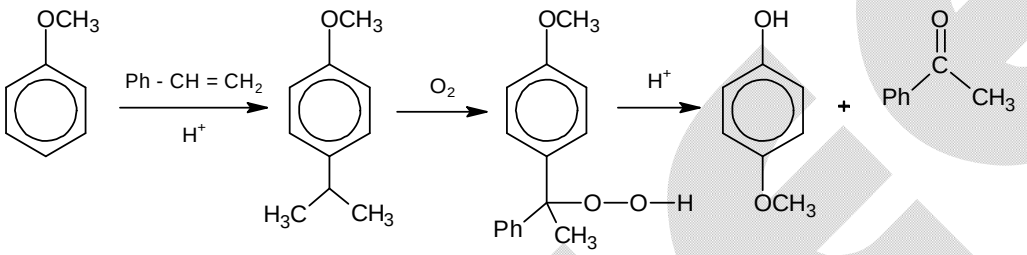
$H_2PO_4^- \rightarrow HPO_4^{2-} + H^+$

NaCl(aq) solution $\left[H^+\right]\left[OH^-\right] = \sqrt{kw}$ $pH = \frac{pkw}{2}$

Ionization decreased with decrease in temperature.

19. BCD

Sol.



20. BC

Sol. Fact.

21. B

Sol. Cell reaction $H_{2(g)} + I_{2(s)} \rightleftharpoons 2H^+_{(aq)} + 2I^-_{(aq)}$

$$E_{cell} = E^0_{cell} - \frac{0.0591}{2}\log\frac{\left[H^+\right]^2\left[I^-\right]^2}{P_{H_2}}$$

$$0.7714 = 0535 - \frac{0.0591}{2}\log\frac{\left(H^+\right)^2(0.1)^2}{1}$$

$$pH = 3$$

22. C

Sol. $NaOH + Zn \longrightarrow H_{2(g)}$

$NaOH + P_4 \longrightarrow PH_3$

$NaOH + NH_4Cl \longrightarrow NH_{3(g)}$

23. A

Sol. for node $\psi^2 = 0$,

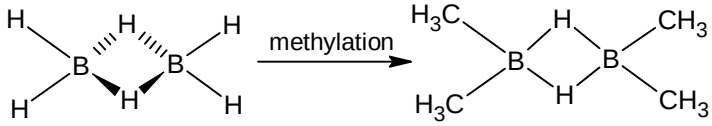
$\therefore \quad 2 - zr = 0$

$$r = \frac{2}{Z}\left(He^+, z = 2\right)$$

$r = 1$

24. D

Sol.



25. B

Sol. $\underset{0.2M}{Al_2(SO_4)_3} \rightarrow \underset{0.2M}{2Al^{3+}} + \underset{0.3M}{3SO_4^{2-}}\ (0.5\,moles/L)$

$\underset{0.1M}{AlPO_4} \rightarrow \underset{0.1M}{Al^{3+}} + \underset{0.1M}{PO_4^{3-}}\ (0.2\,moles/L)$

$Urea \rightarrow 0.1M \quad (0.1\,moles/L)$

$\underset{0.1M}{MgCl_2} \rightarrow \underset{0.1M}{Mg} + \underset{0.2M}{2Cl^-}\ (0.3\,mol/L)$

Colligative properties depends upon no. of particles.

26. A

Sol. $E_n = -\dfrac{kze^2}{2r},\ E_n \propto r^{-1}$

$\dfrac{E_n}{P_n} = \dfrac{1}{2}$

$F_n = \dfrac{V}{2\pi r} \propto z^2$

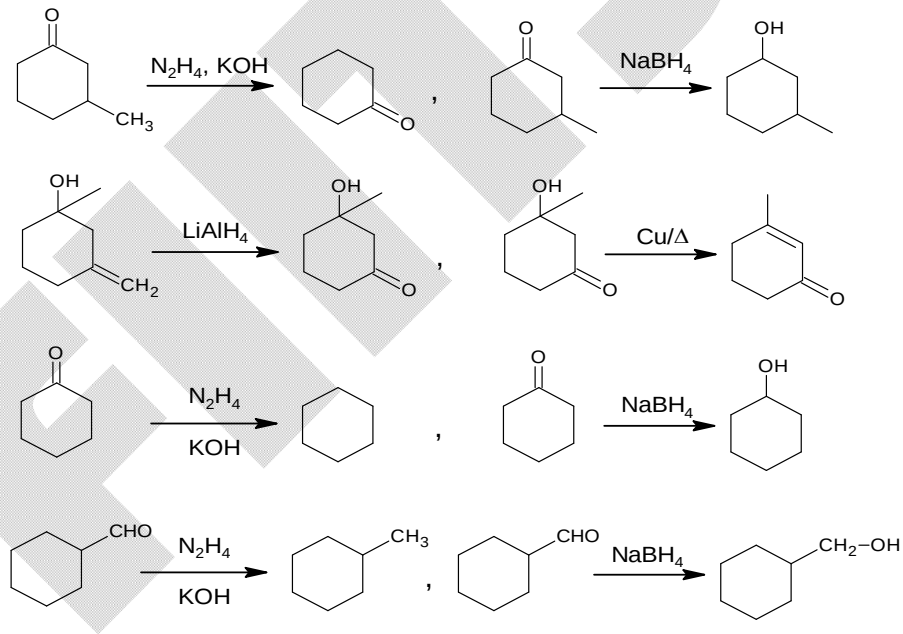
$T_n \times V_n = \dfrac{2\pi r}{V_n} \times V_n,\ t = 1$

27. D

Sol. (P) z = 17, Cl
(Q) z = 36, Kr
(R) z = 26, Fe
(S) z = 63, Eu (third group element)

28. C

Sol.



SECTION – B

29. 7

Sol. The species having odd numbers of electrons and O_2, S_2 and B_2 are paramagnetic.

Paramagnetic $\Rightarrow O_2, S_2, B_2, N_2^{--}(14+2=16=O_2$ configuration), O_2^+, NO and NO_2.

30. 3

Sol. $HF_{(g)} \xrightarrow{H_2O} H^+_{(aq)} + F^-_{(aq)}; \Delta H = ?$

1. $\frac{1}{2}H_2(g) + \frac{1}{2}F_2(g) \longrightarrow HF_{(g)}; \Delta H_1 = -275kJ$

2. $\frac{1}{2}H_{2(g)} \xrightarrow{H_2O} H^+_{(aq)} + e^-; \Delta H_2 = 0kJ$

3. $\frac{1}{2}F_{2(g)} + e^- \xrightarrow{H_2O} F^-_{(aq)}; \Delta H_3 = -335kJ$

From equations (1), (2), (3)

$\Delta H = -335 + 275$

$= -60kJ$

M.wt. of $HF = 1 + 19 = 20$

20 gm of HF $\Rightarrow -60kJ$

1 gm HF $= -3kJ$

31. 3

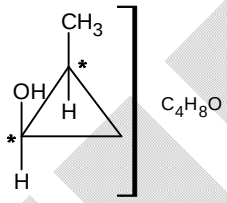
Sol. $NH_3 \Rightarrow 107^o$, $H_2O \Rightarrow 104.5^o$, $CH_4 \Rightarrow 109^o28'$

- PH_3, AsH_3, SbH_3
- H_2S, H_2Se, H_2Fe

] Have bond angle in the range of 90°

32. 4

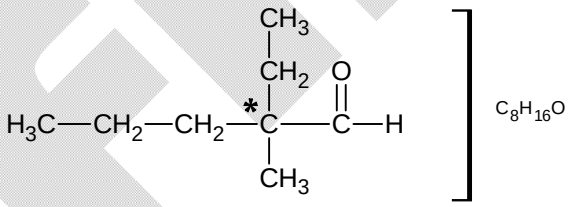
Sol.



C_4H_8O

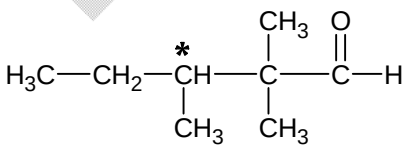
33. 8

Sol.



$C_8H_{16}O$

or



34. 4

Sol. Mole of sodium thiosulphate

$= 2 \times \frac{1}{5.6} = \frac{2}{5.6}$ mols

$O_3 + 2KI \longrightarrow K_2O + I_2 + O_2$

$I_2 + 2Na_2S_2O_3 \longrightarrow Na_2S_4O_6 + 2NaI$

From these equations,

Moles of ozone $= \frac{1}{2} \times$ mols of $Na_2S_2O_3$

$= \frac{1}{2} \times \frac{2}{5.6}$

$= \frac{1}{5.6}$ mole

Volume of ozone $= \frac{1}{5.6} \times 22.4 = 4$

Mathematics

PART – III

SECTION – A

35. ABD

Sol. (A) $P\left(\overline{A}\overline{B}\right)=P\left(\overline{A^c\cap B^c}\right)=P\left(A^c\cap B^c\right)^c=P(A\cup B)P\left(\overline{A\cup B}\right)=P\left(A^c\cup B^c\right)=P(A\cap B)$

$\therefore P(A\cup B)-P(A\cap B)=$ exactly one of either A or B occurs $\Rightarrow$ True

(B) $E=P\left(A\overline{B}\right)+P(A\cap B)+P\left(\overline{A}B\right)+P\left(A^c\cap B^c\right)=$ P(entire sample space) $\Rightarrow$ = True

(C) $P(A\cap B)=0=P(A)\cdot P(B)$ which is not possible if $P(A)>0$ and $P(B)>0$ $\Rightarrow$ False

(D) Since occurrence or non-occurrence of A does not effect the probability of B, hence A and B are independent. $\Rightarrow$ True

36. BD

Sol. Use $R_3\to R_3-\alpha R_1-R_2$ to get $D=\left(ac-b^2\right)\left(a\alpha^2+2b\alpha+c\right)=0$.

37. ABCD

Sol. $f\left(1^+\right)=1=f\left(1^-\right)$ but $f(1)=2;$

$f\left(-1^+\right)=3=f\left(-1^-\right)$

$f\left(2^+\right)=4;f\left(2^-\right)=2;f(2)=4$

38. C

Sol. Let F as $a+bi,a,b\in R,$ where we see from the diagram that $a,b>0$ and $a^2+b^2>1$ (as F lies outside the unit circle)

Since $\alpha=\dfrac{1}{a+bi}=\dfrac{a-bi}{a^2+b^2}=\dfrac{a}{a^2+b^2}-\dfrac{b}{a^2+b^2}i,$

Real part + ve and imaginary part – ve and both less than unity)

We see that the reciprocal of F is in IV quadrant, since the real part is positive and the imaginary part is negative. Also, the magnitude of the reciprocal is:

$$\frac{1}{a^2+b^2}\sqrt{a^2+(-b)^2}=\frac{1}{\sqrt{a^2+b^2}}<1$$

Thus, the only possibility is point C.

39. A

Sol. $$\frac{m\cdot{}^nC_2+n\cdot{}^mC_2}{m\cdot{}^nC_2+n\cdot{}^mC_2+mn}.$$

40. C

Sol. Let $f(x)=y$

$\Rightarrow \quad x=f^{-1}(y)=g(y)\Rightarrow x=e^{e^y}$

$\Rightarrow \quad \dfrac{dx}{dy}=e^{e^y}\cdot e^y=e^{e^y+y}=g'(y)$

Hence, $g'(x)=e^{e^x+x}$

41. B

Sol. $\tan\alpha = -1_1 \text{ and } \tan\beta = -t_2$

Also $t_2 = -t_1 - \dfrac{2}{t_1}$

$t_1t_2 + t_1^2 = -2$

$\tan\alpha\tan\beta + \tan^2\alpha = -2$

$\Rightarrow$ (B)

42. B

Sol. (P) $L_1 : \dfrac{x-1}{1} = \dfrac{y-0}{1} = \dfrac{z-2}{-5}; \vec{V}_1 = \hat{i} + \hat{j} - 5\hat{k}$

$L_2 : \dfrac{x-2}{2} = \dfrac{y-1}{2} = \dfrac{z+3}{-10}; \vec{V}_2 = 2\left(\hat{i} + \hat{j} + -5\hat{k}\right)$

Hence lines are parallel and both contains the points (1, 0, 2) and $(2,1,-3) \Rightarrow$ coincident line both L_1 and L_2 may lie in an infinite number of planes

(Q) $\left.\begin{matrix}\vec{V}_1 = 2\hat{i} + 2\hat{j} - \hat{k} \\ \vec{V}_2 = \hat{i} - 2\hat{j} + 3\hat{k}\end{matrix}\right] \Rightarrow$ Lines not parallel.

Also both intersect at (3, 5, 1).
Hence lines are intersecting hence they lie on a unique plane.

(R) $L_1 : \dfrac{x-0}{-6} = \dfrac{y-1}{9} = \dfrac{z-0}{-3} = t$

$L_2 : \dfrac{x-1}{2} = \dfrac{y-4}{-3} = \dfrac{z-0}{1} = s$

L_1 is parallel to $-6\hat{i} + 9\hat{j} - 3\hat{k}$

L_2 is parallel to $2\hat{i} - 3\hat{j} + \hat{k}$

$\Rightarrow$ Lines parallel but not coincident
As (0, 1,0) does not line on L_2, not intersecting
Hence, L_1, L_2 lies in a unique planes
(S) Lines are skew can be verified.

43. D

Sol. (P) $f(x) = \int x^{\sin x}\left(1 + x\cos x \cdot \ln x + \sin x\right)dx$

$F(x) = x^{\sin x} = e^{\sin x \ln x}$

$\therefore \quad xF'(x) = x^{\sin x}\left(x\cos x \ln x + \sin x\right)$

$\therefore \quad f(x) = \int\left(F(x) + xF'(x)\right) = xF(x) + C$

If $f(x) = x \cdot x^{\sin x} + C$

$f\left(\dfrac{\pi}{2}\right) = \dfrac{\pi}{2} \cdot \dfrac{\pi}{2} + C$

$\Rightarrow \quad C = 0$

$\therefore \quad f(x) = x(x)^{\sin x}; \quad f(\pi) = \pi(\pi)^0 = \pi(\text{irrational})$

(Q) $g(x) = \int \dfrac{\cos x(\cos x + 2) + \sin^2 x}{(\cos x + 2)^2}dx$

$$= \int \underbrace{\cos x}_{II} \cdot \underbrace{\frac{1}{(\cos x + 2)}}_{I} dx + \int \frac{\sin^2 x}{\cos x + 2} dx$$

$$= \frac{1}{\cos x + 2} \cdot \sin x - \int \frac{\sin^2 x}{(\cos x + 2)^2} dx + \int \frac{\sin^2 x}{(\cos x + 2)} dx$$

$$g(x) = \frac{\sin x}{\cos x + 2} + C$$

(R) Let $x + 5 = 14\cos\theta$

$y - 12 = 14\sin\theta$

$\therefore \quad x^2 + y^2 = (14\cos\theta - 5)^2 + (14\sin\theta + 12)^2$

$= 196 + 25 + 144 + 28(12\sin\theta - 5\cos\theta)$

$= 365 + 28(12\sin\theta - 5\cos\theta)$

$\therefore \quad \sqrt{(x^2 + y^2)}\Big|_{min} = \sqrt{365 - 28 \times 13}$

$= \sqrt{365 - 364} = 1$

(S) $k(x) = \int \frac{(x^2 + 1)dx}{(x^3 + 3x + 6)^{1/3}}$

Put $x^3 + 3x + 6 = t^3$

$\Rightarrow \quad 3(x^2 + 1)dx = 3t^2 dt$

$k(x) = \int \frac{t^2 dt}{t} = \frac{t^2}{2} + C$

$k(x) = \frac{1}{2}(x^3 + 3x + 6)^{2/3} + C$

$k(-1) = \frac{1}{2}(2)^{2/3} + C$

$\Rightarrow \quad C = 0$

$\therefore \quad k(x) = \frac{1}{2}(x^3 + 3x + 6)^{2/3}; f(-2) = \frac{1}{2}(-8)^{2/3}$

$= \frac{1}{2}\left[(-2)^3\right]^{2/3} = 2$

44. A

Sol. (P) $T_{r+1} = {}^{15}C_r \cdot (\sqrt{2})^{15-r} \cdot (\sqrt{3})^r$

$= {}^{15}C_r \cdot (2)^{\frac{15-r}{2}} (3)^{r/2}$

If r is even, $(2)^{\frac{15-r}{2}}$ is irrational.

If r is odd, $(3)^{\frac{r}{2}}$ is irrational.

Hence all the 16 terms of the expansion are irrational.

(Q) ${}^nC_1 \cdot 3^1 + {}^nC_2 3^2 + ... + {}^nC_n 3^n = 4095$

$\therefore \quad (1+3)^n = 4096 = 2^{12} = 4^6$

$\Rightarrow \quad n = 6$

(R) ${}^nP_r = {}^nC_r \cdot 5040$

$^nC_r \cdot r! = {}^nC_r \cdot 5040$

$\Rightarrow r! = 5040$

$\Rightarrow r = 7$

(S) $abcde \equiv 2000 = 2^4 \cdot 5^3$

$\equiv 2^3 \cdot 2 \cdot 5 \cdot 5 \cdot 5$

$\equiv 2^2 \cdot 2^2 \cdot 5 \cdot 5 \cdot 5$

Hence, N is of the form 8 2 5 5 5 = $\frac{5!}{3!} = 20$

or 4 4 5 5 5 = $\frac{5!}{2!3!} = 10$

Total = 30

$\Rightarrow$ K = 15

45. A

Sol. (P) For a = 0 $I = \int_0^T \sin^2 x dx = \int_0^T \frac{1-\cos 2x}{2} dx$

$= \left[\frac{x}{2} - \frac{1}{4}\sin 2x\right]_0^T = \frac{T}{2} - \frac{1}{4}\sin 2T$

$\therefore \quad L = \frac{1}{2} - \lim_{t\to\infty} \frac{1}{4}\frac{\sin 2T}{T} = \frac{1}{2}$

$\Rightarrow$ (P) $\to 2$

(Q) For $a = 1, \int_0^T 4\sin^2 x dx \Rightarrow L = 2,$ hence (Q) $\to 4$

(R) For $a = -1, \int_0^T 0 dx = 0 \Rightarrow L = 0$ hence (R) $\to 1$

(S) For $a \neq 0, -1, 1,$

$I = \int_0^T \left(\sin^2 x + \sin^2 ax + 2\sin x \cdot \sin ax\right) dx$

$= \int_0^T \left(\frac{1-\cos 2x}{2} + \frac{1-\cos 2ax}{2} + \cos(a-1)x - \cos(a+1x)\right) dx$

$= \left[x - \frac{1}{4}\sin 2x - \frac{1}{4a}\sin 2ax + \frac{\sin(a-1)x}{a-1} - \frac{\sin(a+1)x}{a+1}\right]_0^T$

$\therefore L = \lim_{T\to\infty} \frac{T}{T} - \lim_{T\to\infty} \frac{1}{T}\left[\frac{1}{4}\sin 2x - \frac{1}{4a}\sin 2ax + \frac{\sin(a-1)x}{a-1} - \frac{\sin(a+1)x}{a+1}\right]_0^T$

$L = 1$ hence (S) $\to 3$

SECTION – B

46. 7

Sol. Equating the components, $3x + 2y + 4z = \alpha x; 2x + 2z = \alpha y$ and $4x + 2y + 3z = \alpha z$

Hence $(3\alpha)x + 2y + 4z = 0$

$2x - \alpha y + 2z = 0$

$4x + 2y + (3 - \alpha)z = 0$

For non-trivial solution $\begin{vmatrix} 3-\alpha & 2 & 4 \\ 2 & -\alpha & 2 \\ 4 & 2 & 3-\alpha \end{vmatrix} = 0$

Use: $C_1 \to C_1 - C_3$ and $C_3 \to C_3 - 2C_2$

$$\begin{vmatrix} -(\alpha+1) & 2 & 0 \\ 0 & 2 & 2(1+\alpha) \\ 1+\alpha & 2 & -(\alpha+1) \end{vmatrix} = 0$$

$$\Rightarrow \quad (\alpha+1)^2 \begin{vmatrix} -1 & 2 & 0 \\ 0 & -\alpha & 2 \\ 1 & 2 & -1 \end{vmatrix} = 0$$

$$\Rightarrow \quad (\alpha+1)^2 \left[-1(\alpha-4)-2(-2)\right] = 0$$

$$\Rightarrow \quad (\alpha+1)^2(\alpha-8) = 0$$

Hence $\alpha = -1$ or 8

47. 7

Sol. Let $I = \int_{-1}^{1} \cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)\left(\cot^{-1}\frac{x}{\sqrt{1-\left(x^2\right)^{|x|}}}\right)dx$ …(i)

Using property

$$I = \int_{-1}^{1} \left(\cot^{-1}\frac{1}{\sqrt{1-x^2}}\right)\left(\cot^{-1}\frac{-x}{\sqrt{1-\left(x^2\right)^{|x|}}}\right)dx \quad \text{…(ii)}$$

On adding

$$2I = \int_{-1}^{1} \cot^{-1}\frac{1}{\sqrt{1-x^2}}\left\{\cot^{-1}\frac{x}{\sqrt{1-\left(x^2\right)^{|x|}}} + \pi - \cot^{-1}\frac{x}{\sqrt{1-\left(x^2\right)^{|x|}}}\right\}dx$$

$$2I = \int_{-1}^{1} \pi\cot^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)dx$$

$$= \int_{-1}^{1} \pi\tan^{-1}\sqrt{1-x^2}\,dx$$

$$= 2\pi\int_{0}^{1} \tan^{-1}\sqrt{1-x^2}\,dx \quad \text{…(iii)}$$

(As $\tan^{-1}\sqrt{1-x^2}$ is even function)

$$\therefore \quad I = \pi\int_{0}^{1} \underbrace{1}_{II}\underbrace{\tan^{-1}\left(\sqrt{1-x^2}\right)dx}_{I} \quad \text{…(iv)}$$

Integrating by parts

$$I = \underbrace{\pi\tan^{-1}\left(\sqrt{1-x^2}\right)\cdot x\Big]_0^1}_{\text{zero}} - \int_0^1 \frac{x}{\left(1+1-x^2\right)}\frac{(-x)}{\sqrt{1+x^2}}dx$$

$$= \pi\int_0^1 \frac{x^2}{\left(2-x^2\right)\sqrt{1-x^2}}dx$$

Let $x = \sin\theta$

$\Rightarrow \quad dx = \cos\theta d\theta$

$$I = \pi\int_0^{\pi/2}\frac{\sin^2\theta}{\left(2-\sin^2\theta\right)}d\theta = -\pi\int_0^{\pi/2}\frac{2-\sin^2\theta-2}{2-\sin^2\theta}d\theta$$

$\therefore \quad I = 2\pi\int_0^{\pi/2}\frac{d\theta}{2-\sin^2\theta}-\frac{\pi^2}{2}$

$$= 2\pi\int_0^{\pi/2}\frac{\sec^2\theta d\theta}{2+2\tan^2\theta-\tan^2\theta}-\frac{\pi^2}{2}$$

$$= 2\pi\int_0^{\pi/2}\frac{\sec^2\theta d\theta}{2+\tan^2\theta}-\frac{\pi^2}{2}$$

Put $\tan\theta = t$

$$I = 2\pi\int_0^{\infty}\frac{dt}{2+t^2}-\frac{\pi^2}{2} = 2\pi\frac{1}{\sqrt{2}}\tan^{-1}\frac{t}{\sqrt{2}}\Big]_0^{\infty}-\frac{\pi^2}{2}$$

$$= \frac{\pi^2}{\sqrt{2}}-\frac{\pi^2}{2} = \frac{\pi^2\left(\sqrt{2}-1\right)}{2} = \frac{\pi^2\left(\sqrt{a}-\sqrt{b}\right)}{\sqrt{c}}$$

$\Rightarrow \quad a = 2, b = 1$ and $c = 4$

$\Rightarrow \quad a+b+c = 2+1+4 = 7$

48. 3

Sol. $(a-1)x^2 = x(2b+3)$

The above equation is satisfied by three distinct values of x therefore it is an identity.

$\therefore \quad 2-2a = 0$

$\Rightarrow \quad a = 1$

and $\quad 2b+3 = 0$

$\Rightarrow \quad b = \frac{-3}{2}$

Now, $\quad f(x) = 2x+1,$

Let $\quad g(x) = px+q$

$\Rightarrow \quad g'(x) = p$

$f(g(x)) = 6x-7$

$\Rightarrow \quad 2(px+q)+1 = 6x-7$

$\Rightarrow \quad 2px+2q+1 = 6x-7$

$\Rightarrow \quad 2p = 6$

$\Rightarrow \quad p = 3$ and $q = -4$

$\therefore \quad g'(2012) = 3$

49. 4

Sol. Clearly, $x^2-4x+\alpha > \text{ or } \geq \frac{-1}{\sqrt{3}}\forall x \in R$

$\Rightarrow \quad x^2-4x+\alpha+\frac{1}{\sqrt{3}} > 0\forall x \in \forall R$

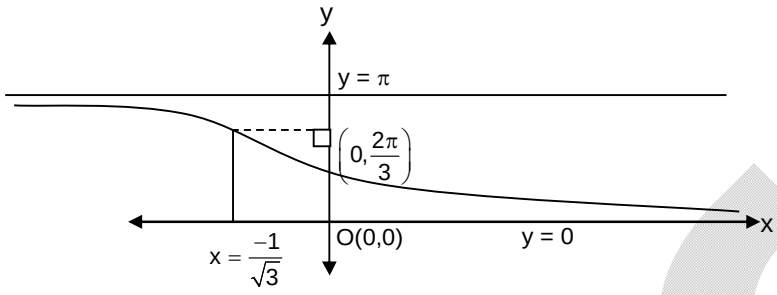
So, discriminant $< 0 \Rightarrow 16-4\left(\alpha+\frac{1}{\sqrt{3}}\right) < 0$

$$4-\alpha-\frac{1}{\sqrt{3}}<0$$

$\Rightarrow \quad \alpha > 4-\frac{1}{\sqrt{3}}$

$\therefore \quad \alpha \in \left(4-\frac{1}{\sqrt{3}},\infty\right)$

Hence minimum integral $\alpha = 4$



Graph of $y = \cot^{-1} x$

50. 9

Sol. Since, ${}^{m}C_1, {}^{m}C_2$ and ${}^{m}C_3$ are the first, third and fifth term of an A.P., so

$2\,{}^{m}C_2 = {}^{m}C_1 + {}^{m}C_3$

$\Rightarrow \quad 2\left(\frac{m!}{2!(m-2)!}\right) = m + \frac{m!}{3!(m-3)!}$

$\Rightarrow \quad m\left(m^2-9m+14\right)=0$

As $m \neq 0$, so $(m-2)(m-7)=0$

$\Rightarrow$ m = 7(as 6th term = 21, so m = 2 is not possible)

Now, $21 = {}^{7}C_5\left(\sqrt{2^{\log(10-3^n)}}\right)^{7-5} \times \left(\sqrt[5]{2^{(n-2)\log 3}}\right)^5$

$\Rightarrow \quad 21-21\left(2^{\log(10-3^n)+\log 3^{(n-2)}}\right)$

$\Rightarrow \quad 1 = 2^{\log(10-3^n)\cdot 3^{n-2}}$

$\Rightarrow \quad 10\cdot 3^{n-2} - 3^{2n-2} = 1$

$\Rightarrow \quad 3^{2n} - 10\cdot 3^n + 9 = 0$

$\Rightarrow \quad \left(3^n-1\right)\left(3^n-9\right)=0$

$\Rightarrow \quad n = 0$ or 2

But $n \in N$, so $n = 2$

Hence, $(m+n) = (7+2) = 9$

51. 5

Sol. $V_2 = 3V_1 - 2V_0 = 5$

$V_3 = 3V_2 - 2V_1 = 9$

$V_4 = 3V_3 - 2V_2 = 17$

$V_5 = 3V_4 - 2V_3 = 33$

$V_6 = 3V_5 - 2V_4 = 65$